

Hybridisation and Thin-Film Sensors Key Enablers of **Flexible Electronics** Today



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Flexible electronics is making its presence felt in almost every field, be it wearable, consumer electronics, medical, industrial or lighting. Still in early development, printed and flexible electronics is also enabling a much-talked-about trend today—the Internet of Things (IoT). Let us take a look at the latest happenings in the area, sneak-peak into the research and development activities and some interesting examples of commercial applications and applications currently under active research.

Large-area and low-cost integration giving rise to real-life applications

In the past year, we continued to see increased interest in the capabilities and technologies related to printed electronics. Printed electronics continues to fit the profile of an emerging technology space as awareness and participation in the area grows. “State-of-the-art capabilities of flexible and printed electronics include logic and memory devices, displays and lighting, thin-film batteries, photovoltaics as well as a multitude of sensors,” says Luisa Petti, PhD student, Wearable Computing Laboratory, Swiss Federal Institute of Technology Zurich.

Petti adds, “Recently, efforts have also moved towards large-area and low-cost integration of all these devices into fully-flexible or stretchable systems.” Therefore more and more real-life system applications are being proposed and demonstrated.

It is important to distinguish between hybrid integration of rigid conventional silicon based electronics with flexible electronics and fully-

flexible printed systems using only low-temperature materials. Petti says, “On one side, the hybrid approach allows taking advantages of the high performance of rigid silicon technology and at the same time expands its applications using flexible electronics technology.”

A few examples of systems developed employing this hybrid approach are LG G Flex mobile phone, Apple Watch and MC10 Biostamp.

Listed below are some other select examples that show the versatility of printed electronics.

Sleep mask for prevention of diabetic disease. Niche applications are starting to become more apparent within a range of high-value market sectors. Development of a healthcare application that utilises plastic electronics is one such example.

PolyPhotonix, based at Centre for Process Innovation (CPI), has developed a light-therapy sleep mask, Noctura 400, for the prevention and treatment of diabetic retinopathy, a disease caused by diabetes. It is one of the most common causes of blindness in the western world, informs Steven Bagshaw, marketing executive, CPI.

Designed as a monitored home based therapy, the sleep mask offers a patient-centric, non-invasive treatment that can be delivered at a fraction of the cost of the current interventions—laser photocoagulation surgery or intraocular drug injection. Bagshaw says, “With 3.5 million diabetes sufferers in Britain, the technology has the potential to save National Health Service (NHS) £1 billion per year upon adoption.”

He adds, “The key message for flexible electronics from the success of PolyPhotonix is that the company identified a game-changing market application where functional benefits of plastic electronics added significant value to the product.

The Noctura 400 Sleep Mask
(Image courtesy: dailymail.co.uk)

